



Potassium Disturbances

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Introduction

Potassium, the major intracellular cation in the human body, is found in abundance in fruit (e.g., bananas, oranges), vegetables (e.g., spinach, tomatoes), fish and meat. The average daily dietary intake is 50–100 mmol. Normal plasma potassium concentration is 3.5–5 mmol/L.

Hypokalemia, defined as a plasma potassium concentration of less than 3.5 mmol/L, can be mild (3–3.5 mmol/L), moderate (2.5–3 mmol/L) or severe (<2.5 mmol/L). *Hyperkalemia*, defined as a plasma concentration of greater than 5 mmol/L, can be mild (5–6 mmol/L), moderate (6.1–6.5 mmol/L) or severe (>6.5 mmol/L). Hypokalemia and hyperkalemia are associated with potentially fatal cardiac arrhythmias. They can also cause musculoskeletal, neurologic and psychiatric complications.

Goals of Therapy

- Prevent life-threatening cardiac arrhythmias and improve neuromuscular conductivity
- Re-establish normal body stores of potassium (K⁺) and prevent undue losses or accumulations
- Identify and correct underlying causes

Investigations

- History with attention to possible etiology (see [Table 1](#))
- Physical examination to assess cardiac rhythm, muscle weakness, paresthesias, blood pressure (consider hyperaldosteronism in the presence of hypokalemia and high blood pressure) and total body sodium (which heavily influences potassium homeostasis)
 - total body sodium is assessed indirectly by examination of the extracellular fluid volume (ECFV). Edema, hypertension, elevation of the jugular venous pressure (JVP), ascites and pleural effusions are features of expansion of the ECFV. Loss of skin turgor, hypotension (especially postural changes) and decreased JVP are features of contraction of the ECFV
- ECG when clinically necessary, e.g., severe hypo- or hyperkalemia
- Laboratory tests may be useful in assessing hypokalemia or hyperkalemia, if the cause not already evident:
 - rule out pseudohyperkalemia (possible with thrombocytosis or severe leukocytosis). In these situations, the longer the blood stands, the greater the rise in serum potassium
 - rule out in vitro hemolysis or forearm contraction;^[1] hemolysis is common and requires that the test be repeated. Taking a blood sample without a tourniquet is a practical next step. Excessive forearm contraction increases venous K⁺ due to repeated muscle depolarization–induced K⁺ efflux exceeding reuptake via the Na⁺/K⁺-ATPase
 - urea, creatinine, Na⁺, K⁺, Cl[−], glucose
 - Mg⁺⁺ in refractory hypokalemia, especially when patient is at risk for hypomagnesemia, e.g., alcohol use disorder; diarrhea; or taking cisplatin, diuretics or proton pump inhibitor
 - arterial or venous blood gases or total CO₂. Systemic pH has profound effects on K⁺ movement in and out of cells
 - urine sample for Na⁺, K⁺, Cl[−] and osmolality to calculate the transtubular K⁺ concentration gradient (TTKG), the potassium-creatinine ratio and, in select cases, 24-hour urine collection for Na⁺, K⁺ and creatinine
 - aldosterone, cortisol, renin, screening tests for adrenal disease, thyroid-stimulating hormone (TSH) for thyrotoxicosis, kidney ultrasound to assess size of kidneys, and renal artery imaging in select cases

Table 1: Causes of Potassium Disturbances

Hyperkalemia	Hypokalemia
Drug-induced: ACE inhibitors, aminocaproic acid, ARBs, beta-blockers, cyclosporine, digoxin overdose, drospirenone, heparin, K ⁺ supplements, K ⁺ -sparing diuretics (especially steroidal and nonsteroidal MRAs), ketoconazole, NSAIDs, penicillin G potassium, pentamidine,	Drug-induced: acetazolamide, aminoglycosides, amphotericin B, antipseudomonal penicillins, beta ₂ -agonists, caffeine, foscarnet, insulin, laxatives, licorice, long-term systemic corticosteroid therapy, loop and thiazide diuretics, theophylline, tocolytic agents

Hyperkalemia	Hypokalemia
succinylcholine, tacrolimus, trimethoprim or TMP/SMX (high-dose or in susceptible patients, i.e., older adults, kidney failure)	Cushing syndrome
	Dialysis/plasmapheresis
Acidosis	Familial history (Bartter or Gitelman syndrome)
Crush injury, trauma, hemolysis, tumor lysis, massive transfusions	Hypomagnesemia
	Inadequate dietary intake
Disease-induced: kidney failure, diabetes, adrenal insufficiency, hyporeninemic hypoaldosteronism	Increased losses: diarrhea, sweat, ileostomy, osmotic diuresis (diabetes), vomiting
Familial history of hyperkalemia	Metabolic alkalosis
Herbal and nutritional supplements (e.g., alfalfa, nettle, noni juice, Siberian ginseng), especially in the presence of kidney failure	Mineralocorticoid excess, e.g., primary aldosteronism
Potassium-containing salt substitutes	Renovascular disease (both large vessel [e.g., renal artery stenosis] and small vessel (e.g., scleroderma)
Abbreviations: ACE = angiotensin converting enzyme; ARB = angiotensin II receptor blocker; MRA = mineralocorticoid receptor antagonist; NSAID = nonsteroidal anti-inflammatory drug; TMP/SMX = trimethoprim/sulfamethoxazole	

Hyperkalemia

Therapeutic Choices

Nonpharmacologic Choices

- Stop K⁺ supplements and/or drugs that induce hyperkalemia (see [Table 1](#)). If necessary, resume K⁺ supplements at a reduced dose once the hyperkalemia is resolved.
- Reduce dietary K⁺ intake to ≤60 mmol/day.

In mild hyperkalemia (plasma K⁺ 5–6 mmol/L), these measures are usually sufficient, reducing K⁺ in a few days. Increasing K⁺ level, ongoing K⁺ absorption, release of intracellular K⁺ (e.g., red blood cell breakdown intravascularly, hematomas, GI bleeding) or kidney failure require further measures, e.g., hemodialysis.

Pharmacologic Choices

Do not rely only on the plasma K⁺ or ECG to determine the urgency of treatment. Though changes in the ECG have an uncertain diagnostic and prognostic significance, they are usually used as immediate evidence of a need for prompt treatment. Cardiac toxicity depends not only on the level of plasma K⁺ but also on the rate of increase, chronicity of hyperkalemia, levels of other electrolytes (hypocalcemia, hyponatremia and acidosis increase cardiotoxicity) and cardiac irritability. Consider all of these factors.

In severe hyperkalemia (K⁺ >6.5 mmol/L) or when significant or advanced ECG changes are present (loss of P waves or widening of QRS complexes), *continuous cardiac monitoring* should accompany treatment. Give **IV calcium** promptly and begin **insulin**. Monitor plasma glucose to avoid overtreating and causing hypoglycemia. Initiate K⁺ removal simultaneously (see [Figure 1](#), [Table 2](#)).

In less severe situations, K⁺ removal with or without redistribution agents (e.g., insulin) may be sufficient. Estimate kidney function and ongoing gain of K⁺ in extracellular fluid. Initiate treatment early and more aggressively when kidney failure is present or there is rapid and severe release of intracellular K⁺ (e.g., rhabdomyolysis, tumor lysis syndrome). Treatment can be less aggressive when there is slow or no input (e.g., hyperkalemia induced by K⁺-sparing diuretics).

Membrane Antagonists

Calcium gluconate or **calcium chloride** antagonize the adverse cardiac effects of K⁺ and should be used in the presence of ECG changes or a high risk of cardiotoxicity since they are fast-acting. Calcium IV has a rapid onset but relatively short duration of action and should be administered simultaneously with other treatments.

Redistribution Agents

These agents act for a longer period than membrane antagonists. Stimulation of insulin and beta receptors as well as correcting acidosis facilitates the reuptake of K⁺ into cells.

When pharmacologic therapy is indicated (see [Figure 1](#)), **insulin** should be the first choice since it is known to be effective and has limited side effects.^[2] For severe hyperkalemia (K⁺ >6.5 mmol/L), insulin must be administered as an IV bolus in conjunction with IV calcium.^[3] Use an insulin infusion if the situation is less urgent. **Dextrose** (50 g per 10 units insulin) is given to avoid hypoglycemia due to the effect of insulin, but do not administer too quickly because an acute increase in plasma tonicity can induce a rise in plasma K⁺. Expect a 1–1.5 mmol/L fall in plasma K⁺ in 60 minutes.

Sodium bicarbonate (NaHCO_3) is usually reserved for hyperkalemia associated with significant metabolic acidosis. It has a synergistic effect with insulin in the presence of mild acidosis.^[4] In the absence of low serum bicarbonate concentration or pH, sodium bicarbonate has a smaller effect.^[5] To avoid an acute increase in plasma K^+ induced by an osmolality change, hypertonic NaHCO_3 solutions should not be used.^[6] The correction of acidosis in hypocalcemic patients may induce tetany. Insulin administration is faster, more reliable and more effective than sodium bicarbonate.^[3]

The β_2 -agonist **salbutamol** is effective in lowering plasma K^+ by increasing activity of membrane-bound Na^+/K^+ -ATPase activity.^[2] Concurrent administration of insulin and salbutamol has a synergistic effect.^[7] High doses of nebulized salbutamol have an effect similar to IV salbutamol;^[8] however, up to 50% of patients with chronic kidney failure are resistant to nebulized salbutamol.^[9] Administration using a metered-dose inhaler with spacer is also effective.^[10] Reserve salbutamol for patients at low risk of coronary artery disease or for life-threatening situations when other therapies have failed, because it is arrhythmogenic and has the potential to exacerbate angina.

Potassium Removal

In patients with volume contraction and oliguria (prekidney disease), volume repletion should be implemented to quickly improve kidney K^+ excretion.

The administration of **loop diuretics** in patients with sufficient kidney function can significantly increase kidney K^+ excretion. Provide enough fluid with the diuretic to avoid volume contraction.

Administration of mineralocorticoids (e.g., **9-alpha-fludrocortisone**) may be necessary when urinary potassium excretion is markedly impaired. Assessing urinary potassium and sodium handling with a spot urine test is important in this situation.

If large amounts of K^+ must be removed rapidly, hemodialysis is the technique of choice. Because time is required to prepare the equipment and to insert a catheter, other treatments must be initiated while preparing for dialysis. Peritoneal dialysis is far less efficient in acutely reducing plasma K^+ . Continuous kidney replacement therapies (CKRT) are not efficient methods for removing K^+ in the short term.

Potassium-Binding Agents

Potassium-binding agents can promote increased K^+ removal. **Cation-exchange resins** (sodium polystyrene sulfonate, calcium polystyrene sulfonate) promote the exchange of Na^+ and Ca^{++} for K^+ , respectively, in the bowel; they also bind calcium and magnesium. **Sodium zirconium cyclosilicate (SZC)** is a nonpolymer compound that exchanges K^+ for sodium and hydrogen ions. **Patiromer** is a polymer that exchanges K^+ for calcium ions. The last 2 agents are more taste-tolerable compared to the resins, but are much more expensive. This should be considered when deciding between potassium binders.

Cation-exchange resins have widespread clinical use with apparent efficacy, but the K^+ -lowering effect of single-dose resin-cathartic therapy is subject to debate.^{[2][11][12][13]} The addition of resins does not seem to increase bowel K^+ removal above the effect of the diarrhea induced by the simultaneous administration

of osmotic or secretory cathartics.^[14] Because cation-exchange resins are constipating, it is suggested that they are given with a **laxative**.

Resins have serious side effects. The Na^+ released in exchange for K^+ may lead to volume overload. Rectal ulceration or colonic necrosis has been described when sodium polystyrene sulfonate mixed in **sorbitol** is given orally or by enema.^[15] Necrosis may be caused by sorbitol rather than by resins.^{[16][17]} The duration of drug contact with the mucosa may be a risk factor. A cleansing enema (sodium-free) is recommended to reduce this risk.

Given the potential problems with cation-exchange resins, some recommend trying all other alternatives for managing hyperkalemia first.^[12] Others feel that resins should continue to be used when indicated, preferably administered in water.^[13] Resins in sorbitol should not be used in the immediate postsurgical period or in patients with compromised GI function.^[13]

SZC and patiromer may also be helpful in the acute treatment of more severe hyperkalemia.^[18]^[19] Evidence suggests SZC has a quicker onset of action and, therefore, may be more preferable in the acute setting. A dose-dependent onset of effect can be seen within the first hour; in contrast, patiromer alone demonstrated a reduction 7 hours after administration, reflecting its longer onset of action.^[20]

In addition to treating acute severe hyperkalemia, cation-exchange resins have been shown to be effective in treating chronic mild hyperkalemia.^[21] Both patiromer and SZC have also been shown to be effective for this indication.^{[22][23]}

SZC has been found to maintain normal serum K^+ in patients with CKD and those receiving renin-angiotensin-aldosterone system (RAAS) inhibitors for up to 1 year. SZC is associated with dose-related mild to moderate edema that can be managed with dose reductions and/or with diuretic therapy.^[22]

Patiromer significantly reduces serum K^+ concentrations and facilitates use of RAAS inhibitors in patients with CKD and/or heart failure who either have or are at risk of hyperkalemia. Gastrointestinal adverse effects (constipation, diarrhea, nausea, abdominal discomfort, flatulence) are the most common adverse effects associated with patiromer.^[24]

Therapeutic Tips

- Focus the treatment of chronic hyperkalemia on the cause or pathophysiological mechanism (see [Table 1](#)). The treatments are similar to those used in acute hyperkalemia (diuretics, NaHCO_3 , potassium-binding agents).
- Closely monitor K^+ after introducing or changing the dose of any medication that could induce hyperkalemia such as ACE inhibitors, ARBs and K^+ -sparing diuretics, particularly in patients at risk of hyperkalemia, e.g., kidney failure, older adults, cardiac insufficiency. Monitoring is even more important when a combination of these medications is prescribed (ACE inhibitor or ARB + spironolactone). In this case, measure K^+ 3 and 7 days after the introduction of these medications and after any change in the

dose.^[24] In addition, measure creatinine and K⁺ at 1 week, 4 weeks and, if stable, every 6 months thereafter.^[25]

Hypokalemia

Therapeutic Choices

Nonpharmacologic Choices

- If clinically appropriate, reduce or stop medication that leads to K⁺ loss.
- Determine and treat the etiology (see [Table 1](#), [Figure 2](#)). Assess dietary intake.
- If the deficit is mild (plasma K⁺ 3–3.5 mmol/L) and there are no ongoing losses or clinical conditions warranting prompt treatment, dietary intake of potassium-rich foods should be adequate. If there are still unusual losses, K⁺ supplements or K⁺-sparing diuretics will be needed.

Pharmacologic Choices

The appropriate pharmacologic approach is determined by:

- Relative urgency for treatment.
- The estimated total body deficit for an adult with plasma K⁺ of 3 mmol/L is approximately 200–400 mmol and for plasma K⁺ of 2 mmol/L, approximately 500–700 mmol. *Note:* The true deficit will be larger with an extracellular shift of K⁺ (e.g., acidosis or insulin deficit) and smaller with an intracellular shift of K⁺ (e.g., periodic paralysis or hyperadrenergic state).
- Ongoing losses must be added to the deficit when replacement therapy is planned. Kidney losses can be estimated based on urine K⁺ levels and the volume excreted per hour.
- In the presence of kidney failure, the treatment should be more cautious.

Potassium Salts

In most cases, **potassium chloride** (KCl) is the salt of choice (see [Table 3](#)), and oral administration is the preferred route. If there is no paralytic ileus or suspected absorption problem, oral administration of KCl can rapidly increase plasma K⁺ (40–60 mmol of a liquid preparation will increase plasma K⁺ by 1–1.5 mmol/L).

Reserve the alkalinizing salt **potassium citrate** for hypokalemic patients with metabolic acidosis (e.g., renal tubular acidosis, diarrhea). **Potassium phosphate** is used when severe hypophosphatemia is present.

Reserve IV administration of K⁺ for patients requiring urgent treatment (e.g., respiratory muscle weakness, cardiac arrhythmias especially if digitalis is present, hepatic encephalopathy) or those unable to take oral supplements (e.g., postsurgery, paralytic ileus). To avoid sclerosis, IV K⁺ should be administered via a large peripheral vein at a maximum concentration of 40–60 mmol/L. Administer higher concentrations via a central line with the catheter positioned away from the right atrium or ventricle to avoid direct effects on cardiac membrane potential. In patients with severe hypokalemia, administer K⁺ in a dextrose-free solution to avoid stimulating insulin secretion and subsequent intracellular K⁺ shift.^[26] Administering K⁺ IV rather than PO is more likely to result in hyperkalemia. Thrombophlebitis and pain at site of injection are other risks.

The *rate of administration* depends on the urgency to treat (see [Table 3](#)). Protocols for the treatment of hypokalemia in the hospital setting have been proposed.^{[27][28]}

Potassium-Sparing Diuretics

If increased kidney K⁺ losses are involved in the pathogenesis of hypokalemia (e.g., hyperaldosteronism, concomitant use of other diuretics), **potassium-sparing diuretics** (see [Table 4](#)) may be used to decrease these losses; they also prevent or decrease magnesium losses. **Amiloride**, steroidal mineralocorticoid receptor antagonists (MRAs), (**spironolactone** and **eplerenone**) and **triamterene** (triamterene is available only in combination with hydrochlorothiazide in Canada) are equally effective but differ in side effects. The non-steroidal MRA **finerenone** may also be effective. Use an MRA in patients with primary or secondary hyperaldosteronism. The most frequent and serious side effect is hyperkalemia.

Avoid K⁺-sparing diuretics in patients with kidney or adrenal insufficiency, older adults, patients with diabetes and patients taking other drugs that may increase plasma K⁺ (see [Table 1](#)).

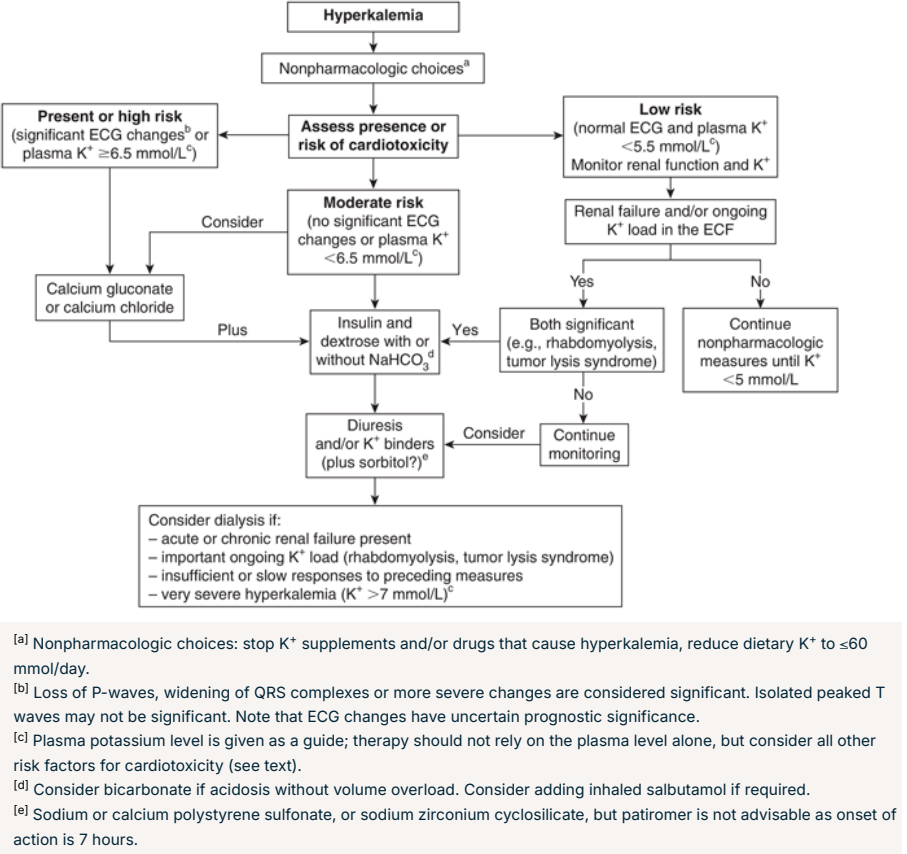
Therapeutic Tips

- The goal in treatment of hypokalemia is to correct total body potassium homeostasis.
- Assess dietary potassium intake.
- Avoid the use of K⁺ supplements and potassium-sparing diuretics together, as the risk of hyperkalemia is greatly increased. Combined use may be required temporarily at the beginning of replacement therapy if kidney K⁺ losses are very high; however, frequent monitoring of plasma K⁺ is mandatory, and one of the drugs should be stopped when plasma K⁺ reaches 3–3.5 mmol/L.
- Treat hypomagnesemia if present.
- If treating acute symptomatic hyponatremia, be cautious of administration of KCl as it will rapidly correct hyponatremia, which can result in serious neurological complications (i.e., osmotic demyelination). Both Na⁺ and K⁺ administration increase total body solute. Hyponatremia is an abnormality of total body tonicity; therefore, both cations and their obligate accompanying anions have equivalent effects on body tonicity.

- Where there is severe metabolic acidosis, especially diabetic ketoacidosis, correct potassium depletion before administering agents such as insulin or bicarbonate as these agents can worsen hypokalemia.

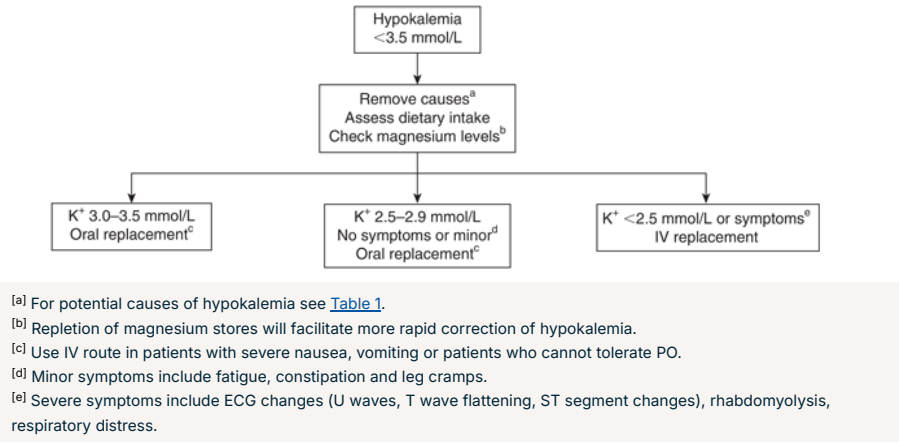
Algorithms

Figure 1: Management of Hyperkalemia



Abbreviations: ECF = extracellular fluid; ECG = electrocardiogram; NaHCO₃ = sodium bicarbonate

Figure 2: Management of Hypokalemia



Drug Tables

Table 2: Drugs Used in the Treatment of Hyperkalemia

Drug/Cost ^[a]	Dosage	Comments	
Drug Class: Beta ₂ -Adrenergic Agonists			
salbutamol Ventolin HFA, generics	10–20 mg by nebulizer 1200 mcg (12 puffs) by pMDI with spacer	Reserved for life-threatening cases when other treatments have failed; risk of arrhythmia or angina; variable response.	
<\$5	Onset: 30 min Duration of action: 2–4 h		
Drug Class: Loop Diuretics			

Drug/Cost ^[a]	Dosage	Comments
furosemide  Lasix, generics <\$5	40–250 mg PO/IV depending on kidney function Onset: 30–60 min Duration of action: to end of increased diuresis (about 4–6 h)	Risk of volume depletion; transient ototoxicity with high-dose furosemide.
Drug Class: Electrolytes		
calcium chloride generics \$10–15	5–10 mL of a 10% solution IV over 2–5 min; may repeat once after 5 min (depending on ECG) Onset: 1–3 min Duration of action: 30–60 min	Continuous ECG monitoring required; infuse more slowly in patients taking digoxin because of increased risk of digoxin toxicity; incompatible with NaHCO ₃ -containing solutions (precipitation). Recommended in cardiac arrest in the presence of hyperkalemia or hypocalcemia or magnesium toxicity. May cause tissue necrosis if extravasation occurs. 1 g calcium chloride = 270 mg elemental calcium = 14 mEq elemental calcium = 7 mmol elemental calcium.
calcium gluconate generics \$15–20	10 mL of a 10% solution IV over 2–5 min; may repeat once after 5 min (depending on ECG) Onset: 1–3 min Duration of action: 30–60 min	Continuous ECG monitoring required; infuse more slowly in patients taking digoxin because of increased risk of digoxin toxicity; incompatible with NaHCO ₃ -containing solutions (precipitation). 1 g calcium gluconate = 93 mg elemental calcium = 4.6 mEq elemental calcium = 2.3 mmol elemental calcium.
sodium bicarbonate \$15–20	50–100 mmol IV over 5 min; repeat Q10–15 min (depending on ECG) Onset: variable, within 1 h	Variable response; risk of tetany if hypocalcemia present (give calcium first); watch for Na ⁺ overload. Can induce bicarbonaturia with an increase

Drug/Cost ^[a]	Dosage	Comments
	Duration of action: 2 h	in renal K ⁺ excretion. Hypertonic NaHCO ₃ solutions should not be used. 8.4% solution is a 1 molar solution.
Drug Class: Insulin		
insulin plus dextrose ~\$40 with 10 mL vial of insulin	Regular insulin 5–10 units IV followed by dextrose 25–50 g over 5 min. If less urgent, infuse 10 units insulin in 500 mL dextrose 10% at the rate of 50–100 mL/h Onset: 30 min Duration of action: 4–6 h	The most reliable medication for redistribution of K ⁺ . Risk of hypoglycemia. Follow plasma glucose. Do not administer dextrose over less than 5 min because acute increase in plasma tonicity can increase K ⁺ in plasma.
Drug Class: Potassium-Binding Agents		
calcium polystyrene sulfonate Resonium Calcium, generics \$5–10	Oral: 15 g 3 or 4 times daily. May mix resin into a paste with sweetened vehicle. Do not mix with orange juice or other fruit juices with high potassium content Rectal: 30 g in 100 mL methylcellulose 2% plus 100 mL of water as a daily retention enema; if possible, enema should be retained for 9 h	Effectiveness seriously questioned. Laxative alone might be as effective. Constipating; risk of bowel ulceration or necrosis. Cleansing enema before PR use recommended. Cleansing enema after PR use to be given after evacuation of the resins or after retention for 1–6 h. Use if sodium overload is a concern. Na content <1 mg/g. Avoid concomitant administration of sorbitol with calcium polystyrene sulfonate as use has been associated with colonic necrosis and intestinal obstruction.
sodium polystyrene sulfonate Kayexalate, Solystat , Odan-Sodium	Oral: 15–60 g in 50–200 mL of 20% sorbitol or water Q4–6H PRN; usual dose: 30 g. Do not mix with orange juice or other fruit juices with high potassium content	Effectiveness seriously questioned. Laxative alone might be as effective. Constipating; risk of bowel ulceration or necrosis.

Drug/Cost ^[a]	Dosage	Comments
Polystyrene Sulfonate, other generics	Onset: 1–2 h	Cleansing enema before PR use recommended. Cleansing enema after PR use to be given after evacuation of the resins or after retention for 1–6 h.
<\$5	Rectal: 30–50 g in 100–200 mL of water or 10% dextrose Q4–6H PRN; retain at least 30–60 min	Do not use if sodium overload is a concern.
	Onset: 30–60 min	Na content 100 mg/g.
	Duration of action: 4–6 h	
sodium zirconium cyclosilicate Lokelma	Acute treatment: 10 g TID PO	Mix dose with 250 mL water.
	Maintenance: 5 g every other day to 10 g daily PO	GI upset and edema possible.
~\$30/10 g sachet		Na content 80 mg/g.
patiromer Veltassa	8.4 g daily PO with a meal initially, then titrate to maximum dose of 25.2 g daily PO based on response	Mix with dose with 90 mL water.
\$10–15		Store in refrigerator and use within 3 months upon removal from refrigerator.
		Mild to moderate constipation, flatulence, hypomagnesemia possible.

^[a] Cost per dose or smallest available pack size, unless otherwise specified; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: ECG = electrocardiogram; GI = gastrointestinal; NaHCO₃ = sodium bicarbonate; pMDI = pressurized metered dose inhaler

Table 3: Potassium Supplements Used in the Management of Hypokalemia

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Comments
Drug Class: Potassium Supplements, intravenous				
potassium chloride, intravenous  generics	Urgent (immediate treatment required): Severe hypokalemia (plasma K ⁺ <2.5 mmol/L)	Urgent (immediate treatment required): 20–40 mmol IV in the first h with continuous ECG	Sclerosis: Administer via a large peripheral vein at a maximum concentration of 40–60 mmol/L.	Doses >40 mmol/h have been given in life-threatening hypokalemia. [31] Potassium (20–40 mmol) can be diluted in 100 mL of NaCl 0.9% if administered via a central vein. A

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Comments
	Symptomatic hypokalemia (respiratory muscle weakness or paresis, paralysis)	monitoring and frequent serum K ⁺ measurements to adjust further rate of administration.	Cardiac arrhythmias and other effects if infusion too rapid.	maximum concentration of 10 mmol in 100 mL of NaCl 0.9% can be administered in a large peripheral vein; however, it could induce pain and sclerosis of the vein.
	Cardiac arrhythmia or conduction disturbances	When plasma K ⁺ = 3 mmol/L, correct the remaining deficit more slowly		Measure serum K ⁺ frequently when a large dose is administered. Measure each time after the administration of 20–40 mmol.
	Less urgent (prompt treatment required):	Less urgent (prompt treatment required):		
	Plasma K ⁺ = 2.5–3 mmol/L	10–20 mmol IV over 1 h (increase serum K ⁺ by 0.25–0.5 mmol/L) ^[29] [30] with ECG monitoring if		
	Hypokalemia with digitalis toxicity, myocardial infarction or ischemia	>10 mmol/h.		
	Hypokalemia with diabetic ketoacidosis (risk of insulin-induced life-threatening hypokalemia)	Should be repeated according to resulting serum K ⁺ . Correct the remaining deficit more slowly		
	Hypokalemia with hepatic insufficiency (risk of hepatic encephalopathy)			
potassium phosphate 	Use when potassium and	Dose according to phosphate	Sclerosis, cardiac	Must be diluted in a compatible IV fluid prior to

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Comments
intravenous generics \$10–20 /dose	phosphate replacement required	requirement in 7.5 mmol increments Dilute dose in IV fluid and administer over 4–12 h Maximum single dose: 45 mmol phosphate (66 mmol K ⁺)	arrhythmias, hypocalcemia.	administration. 1 mmol phosphate administered with approximately 1.5 mmol K ⁺ . Monitor serum K ⁺ and phosphate.

Drug Class: Potassium Supplements, oral

potassium chloride, liquid or powder generics <\$10–20	Not urgent: Plasma K ⁺ >3–3.5 mmol/L	Initially, 40–60 mmol/day PO (divided 2–4 times daily), is usually sufficient	Unpleasant taste, aftertaste, nausea, heartburn.	Salt of choice, especially in alkalotic patients. Rapid absorption, good bioavailability. Avoid use with potassium-sparing diuretics because of increased risk of severe hyperkalemia. If combination therapy is required, frequent monitoring (at least every day) of plasma K ⁺ is mandatory; discontinue either the potassium or the diuretic when plasma K ⁺ reaches 3–3.5 mmol/L.
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potassium chloride, micro-encapsulated Micro-K Extencaps, generics	Not urgent: Plasma K ⁺ >3–3.5 mmol/L	Initially, 40–60 mmol/day PO (divided doses), is usually sufficient	May be less ulceration than with wax matrix.	Salt of choice, especially in alkalotic patients. Rapid absorption, good bioavailability. Avoid use with potassium-sparing diuretics because of increased risk of severe hyperkalemia. If
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Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Comments
<\$10				combination therapy is required, frequent monitoring (at least every day) of plasma K ⁺ is mandatory; discontinue either the potassium or the diuretic when plasma K ⁺ reaches 3–3.5 mmol/L. Avoid in patients with delayed GI transit or impaired esophageal or intestinal motility.

potassium chloride, wax matrix tablet generics <\$10	Not urgent: Plasma K ⁺ >3–3.5 mmol/L	Initially, 40–60 mmol/day PO (divided doses), is usually sufficient	GI symptoms, though less frequent than with liquid, GI ulceration (rare).	Salt of choice, especially in alkalotic patients. Rapid absorption, good bioavailability. Avoid use with potassium-sparing diuretics because of increased risk of severe hyperkalemia. If combination therapy is required, frequent monitoring (at least every day) of plasma K ⁺ is mandatory; discontinue either the potassium or the diuretic when plasma K ⁺ reaches 3–3.5 mmol/L. Avoid in patients with delayed GI transit or impaired esophageal or intestinal motility. Empty wax matrix may appear in stool.
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potassium citrate, effervescent tablet	Not urgent: Plasma K ⁺ >3–3.5 mmol/L	Initially, 40–60 mmol/day PO (divided doses),	Unpleasant taste, aftertaste,	Avoid use with potassium-sparing diuretics because of increased risk of severe hyperkalemia. If
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Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Comments
K-Lyte \$10–20		is usually sufficient	nausea, heartburn.	combination therapy is required, frequent monitoring (at least every day) of plasma K ⁺ is mandatory; discontinue either the potassium or the diuretic when plasma K ⁺ reaches 3–3.5 mmol/L. Useful for patients with metabolic acidosis. Useful for hypokalemia secondary to thiazides given for kidney stones. Increases urinary citrate excretion.
potassium citrate, liquid or tablet ⓘ K-Citra, K-Citra Solution, generics \$10–20	Not urgent: Plasma K ⁺ >3–3.5 mmol/L	Initially, 40–60 mmol/day PO (divided doses), is usually sufficient	Unpleasant taste, aftertaste, nausea, heartburn.	Avoid use with potassium-sparing diuretics because of increased risk of severe hyperkalemia. If combination therapy is required, frequent monitoring (at least every day) of plasma K ⁺ is mandatory; discontinue either the potassium or the diuretic when plasma K ⁺ reaches 3–3.5 mmol/L. Useful for patients with metabolic acidosis. Useful for hypokalemia secondary to thiazides given for kidney stones. Increases urinary citrate excretion.
potassium gluconate ⓘ generics <\$10	Not urgent: Plasma K ⁺ >3–3.5 mmol/L	Initially, 40–60 mmol/day PO (divided doses), is usually sufficient	Unpleasant taste, aftertaste, nausea, heartburn.	Avoid use with potassium-sparing diuretics because of increased risk of severe hyperkalemia. If combination therapy is required, frequent

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Comments
				monitoring (at least every day) of plasma K ⁺ is mandatory; discontinue either the potassium or the diuretic when plasma K ⁺ reaches 3–3.5 mmol/L. Useful in patients with acidosis.

^[a] Cost of 30-day supply of a 20 mmol/day dose unless otherwise specified; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: ECG = electrocardiogram; GI = gastrointestinal

Table 4: Potassium-Sparing Diuretics Used in the Management of Hypokalemia				
Drug/Cost ^[a]	Dosage		Adverse Effects	Drug Interactions
Drug Class: Diuretics, potassium-sparing				
amiloride  Midamor <\$20–40	5–20 mg/day PO		Hyperkalemia, muscle cramps, headaches, GI symptoms (rare).	Avoid NSAIDs , ACE inhibitors, angiotensin II receptor blockers, K ⁺ supplements: may cause severe hyperkalemia.
eplerenone  Inspra, generics \$40–60	25–50mg/day PO daily		Hyperkalemia, dizziness, headaches, rise in serum creatinine.	Avoid NSAIDs , ACE inhibitors, angiotensin II receptor blockers, K ⁺ supplements: may cause severe hyperkalemia.
spironolactone  Aldactone, generics <\$20	25–200 mg/day PO (in single or divided doses). Up to 400 mg/day in patients with hyperaldosteronism		Hyperkalemia, gynecomastia, androgen-like side effects, breast tenderness, GI symptoms.	Avoid NSAIDs , ACE inhibitors, angiotensin II receptor blockers, K ⁺ supplements: may cause severe hyperkalemia.

^[a] Cost of 30-day supply; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations: ACE = angiotensin converting enzyme; GI = gastrointestinal; NSAID = nonsteroidal anti-inflammatory agent

Suggested Readings

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